Suppose you decide to organize a Snakes and Ladders competition with n participants. In this competition, each game of Snakes and Ladders involves three players. After the game is finished, they are ranked first, second, and third. Each player may be involved in any (non-negative) number of games, and the number need not be equal among players.

At the end of the competition, m games have been played. You realize that you forgot to implement a proper rating system, and therefore decide to produce the overall ranking of all n players as you see fit. However, to avoid being too suspicious, if player A ranked better than player B in at least one game, then A must rank better than B in the overall ranking.

You are given the list of players and their rankings in each of the m games. Describe and analyze an algorithm that produces an overall ranking of the n players that is consistent with the individual game rankings, or correctly reports that no such ranking exists.

You must give your algorithm as a reduction to a problem on graphs. Describe your algorithm by answering the following questions:

- What are the vertices? What does each vertex represent?

- What are the edges? Are they directed or undirected?

- If the vertices and/or edges have associated values, what are they?

- **What is the precise relationship between your graph and the stated problem?**

- What standard graph algorithm are you using to solve the problem?

- What is the running time of your entire algorithm, *including* the time to build the graph, *as a function of the original input parameters*?